

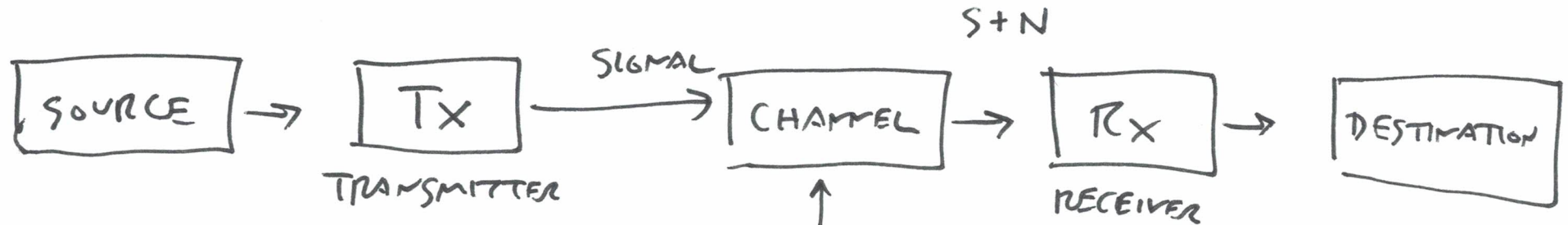
PROBABILITY

THE STUDY OF RANDOM EXPERIMENTS

APPLICATIONS:

- GAMES OF CHANCE, GAMBLING
- FINANCIAL MARKETS
- RELIABILITY TESTING
- SIMULATED SYSTEMS
- ACTUARIAL SCIENCE
- SIGNAL AND IMAGE PROCESSING
- COMMUNICATIONS AND INFO THEORY / NETWORKING
- MACHINE LEARNING.

IN MORE DETAIL,



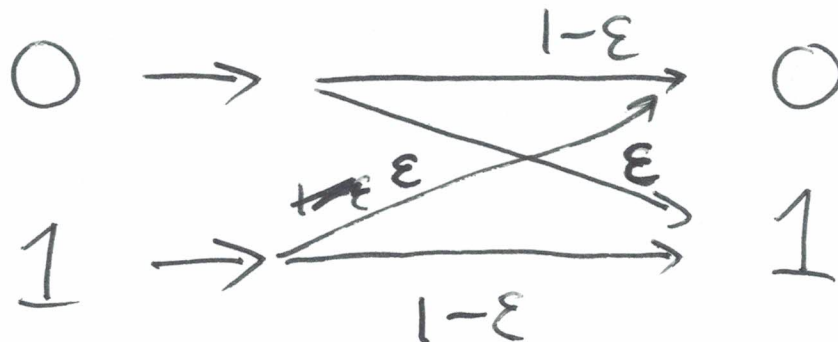
ATMOSPHERIC INTERFERENCE

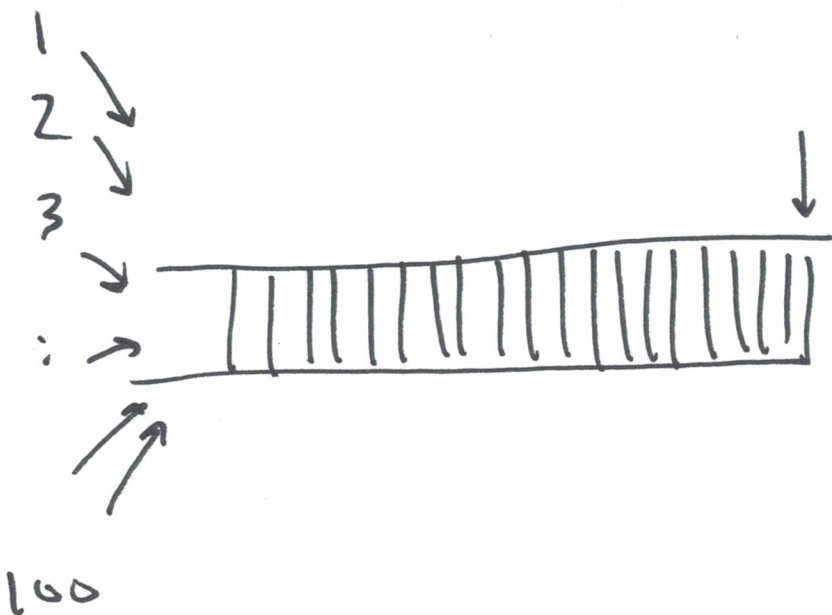
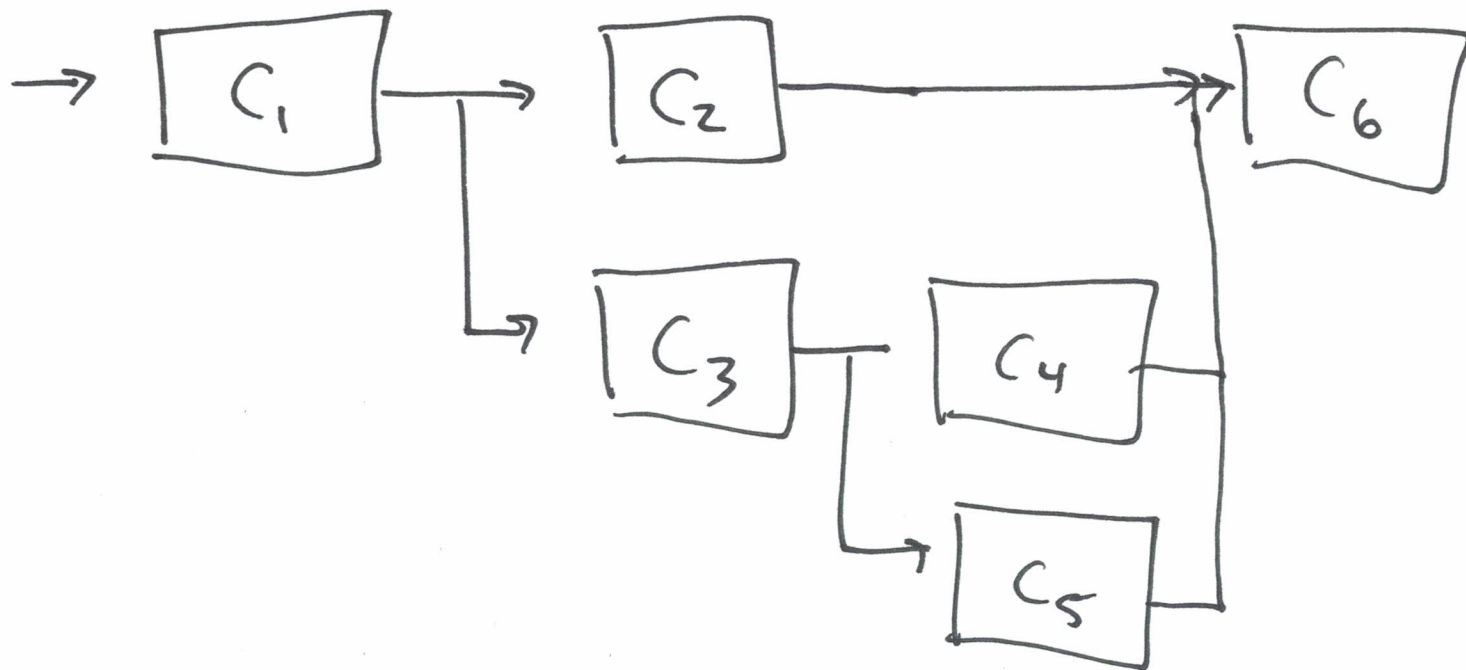
THERMAL NOISE

ACOUSTIC NOISE / CROSSTALK

QUANTIZATION / COMPRESSION

FILM GRAIN





"PICTURE OF A CAT"



$P \left(\left[\text{Cat Icon} \right] \right) > \text{HIGH}$

KEY PROBABILITY TERMS:

AN EXPERIMENT SPECIFIES A PROCEDURE
TO GET A RESULT (OUTCOME).

THE SET OF ALL POSSIBLE OUTCOMES
IS CALLED THE SAMPLE SPACE.

EXACTLY ONE OUTCOME FROM THE
SAMPLE SPACE MUST OCCUR.

$\{0, 1, 2, \dots, 7\} \leftarrow 8 \text{ outcomes}$

$\{(0, 0), (0, 1), \dots, (0, 7),$
 $(1, 0), \dots$

$\leftarrow 64$
OUTCOMES.

$(7, 7)\}$

$(0, 7), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2)$
 $(6, 1), (7, 0)$

$\{0, 1, 2, 3, \dots\}$

THESE ARE ALL DISCRETE SAMPLE
SPACES (COULD BE FINITE OR
INFINITE OUTCOMES)

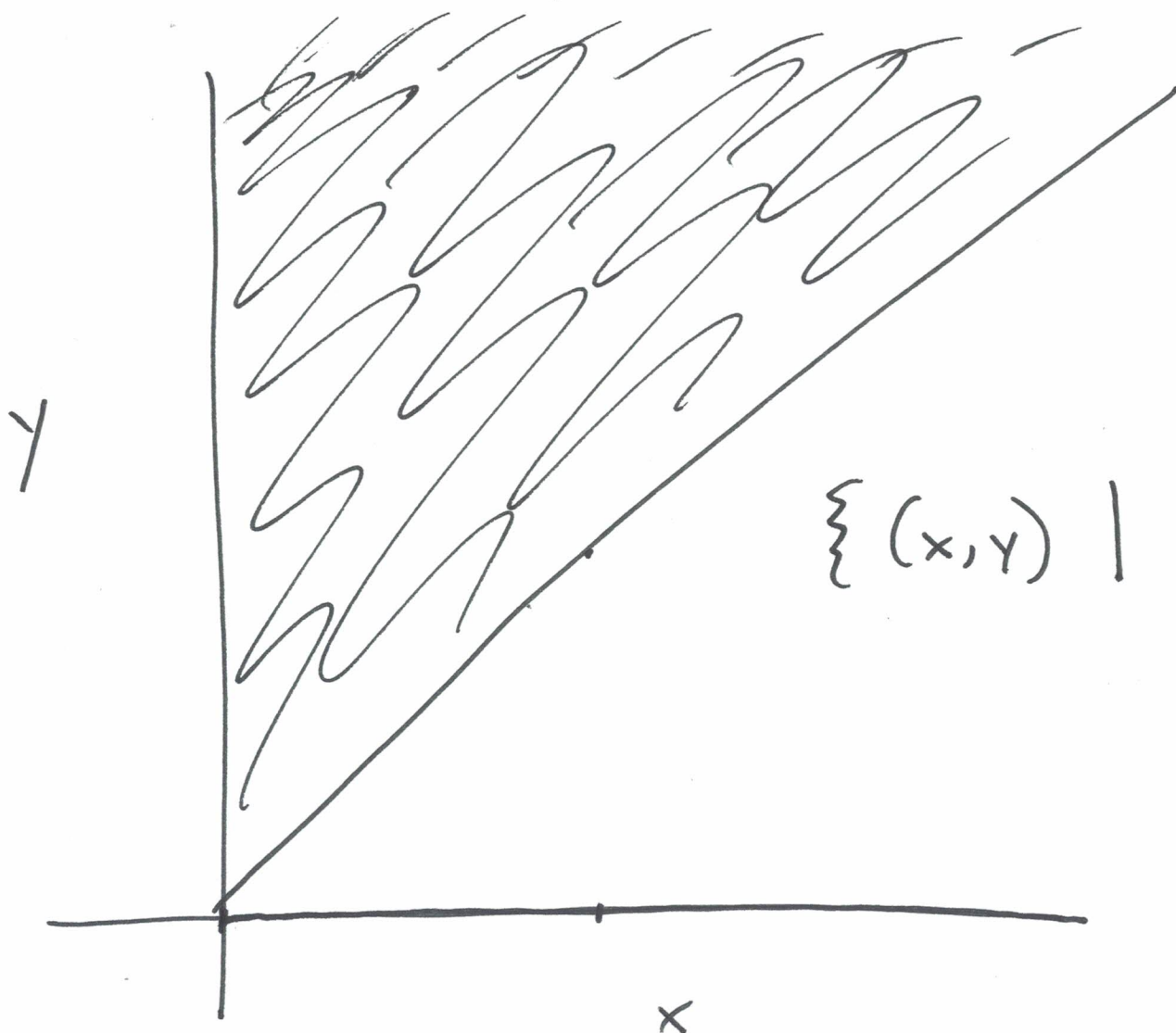
Vs. CONTINUOUS SAMPLE SPACE

TIME UNTIL FIRST CAT:

$[0, \infty)$

OUTCOMES:
1.5
2.83
100.9

(TIME TO FIRST CAT, TIME TO SECOND CAT)



AN EVENT IS A SUBSET OF OUTCOMES.

$$P(\{\text{ODD \# OF CATS}\})$$

$$= P(\{1, 3, 5, 7\})$$

$$\{0\}$$

$$\{1\}$$

$$\{1, 2\}$$

$$\{1, 2, 3\}$$

$$\{2\}$$

⋮

$$\{6, 7\}$$

$$\{7\}$$

$\emptyset$
↑
 $(\emptyset)$
= 0

...

S

$$\{0, 1, 2, 3, 4, 5, 6, 7\}$$

↑

$$P(S) = 1$$

$$S = \{1, 2, 3\}$$

ALL POSSIBLE EVENTS:

$\emptyset$

$$\{1\}$$

$$\{1, 2\}$$

$$\{2\}$$

$$\{2, 3\}$$

$$\{1, 2, 3\}$$

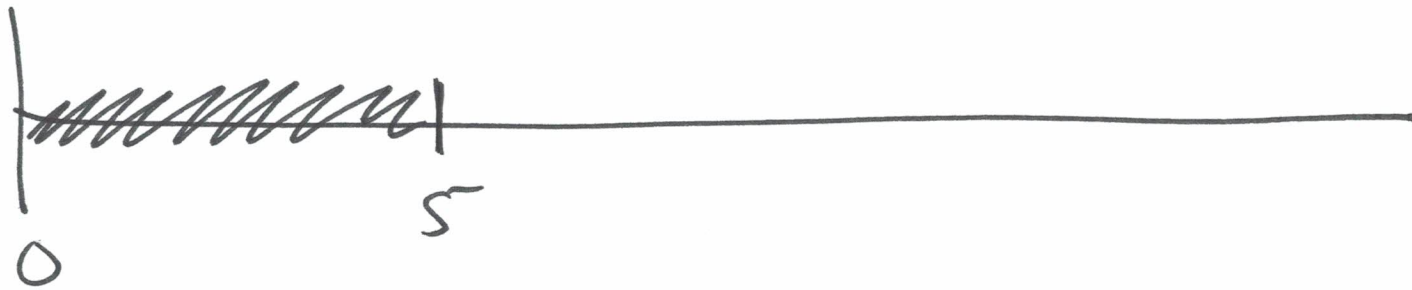
$$\{3\}$$

$$\{1, 3\}$$

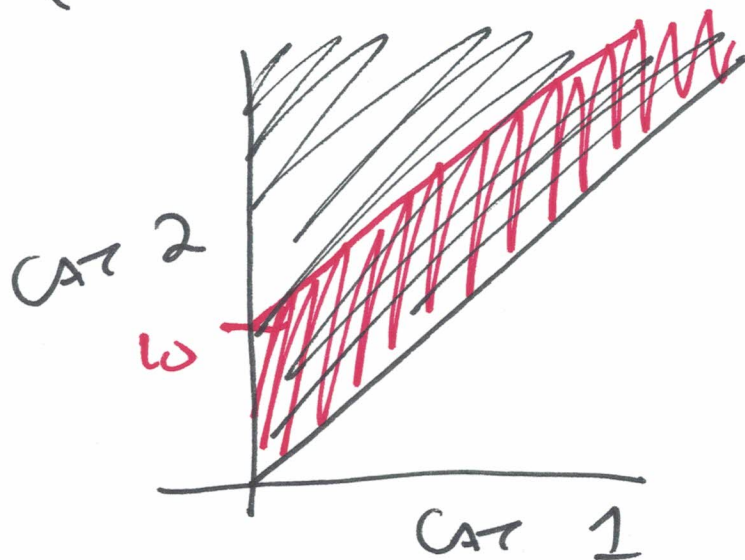
$$8 \text{ TOTAL} = 2^{\boxed{3}}$$

EVENTS IN CONTINUOUS SAMPLE SPACES:

$P(\text{SEE CAT IN FIRST 5 MINUTES})$



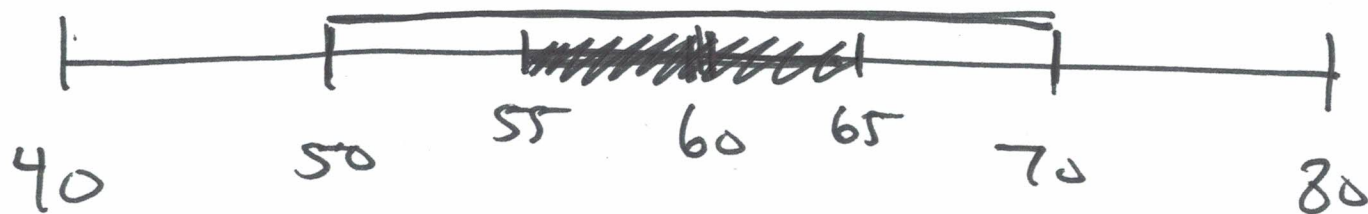
$P(\text{SEE 2 CATS WITHIN 10 MINUTES})$



LENGTH OF CLASS.

SAMPLE SPACE: $[40, 80]$
(CONTINUOUS)

$$P(\text{CLASS LASTS EXACTLY 60 MINS}) = \phi ?$$



WE NEED TO CONSISTENTLY ASSIGN
PROBABILITIES TO EVENTS IN A
WAY THAT MAKES SENSE.

EVERYTHING CAN BE DERIVED FROM 3
BASIC AXIOMS:

1) $P(\text{EVENT}) \geq 0$ (NO NEGATIVE PROBABILITIES!)

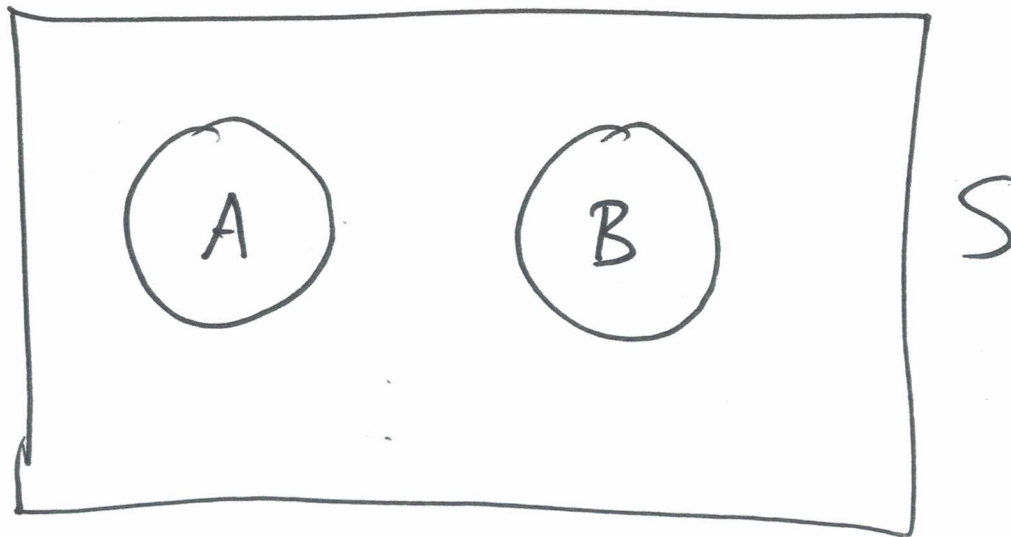
2) $P(S) = 1$ (SOMETHING MUST HAPPEN!
1 UNIT OF TOTAL PROBABILITY)
SAMPLE SPACE

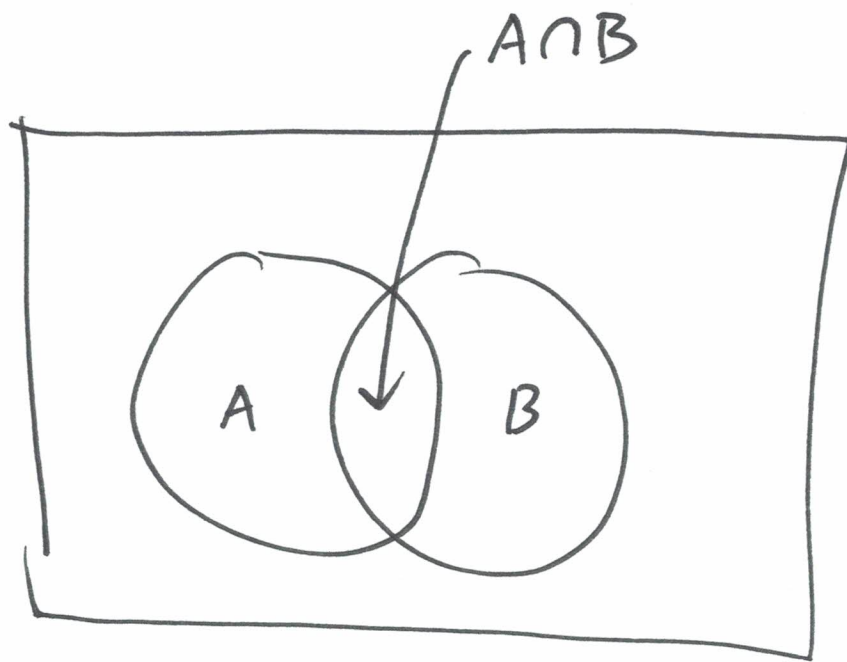
3) IF $A \cap B = \emptyset$

EVENTS A AND B
HAVE NO OUTCOMES
IN COMMON (DISJOINT)

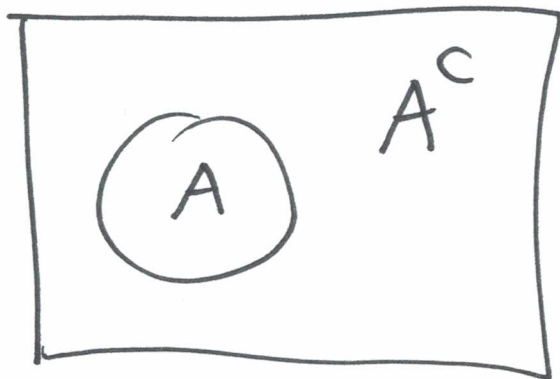
THEN

$$P(A \cup B) = P(A) + P(B)$$

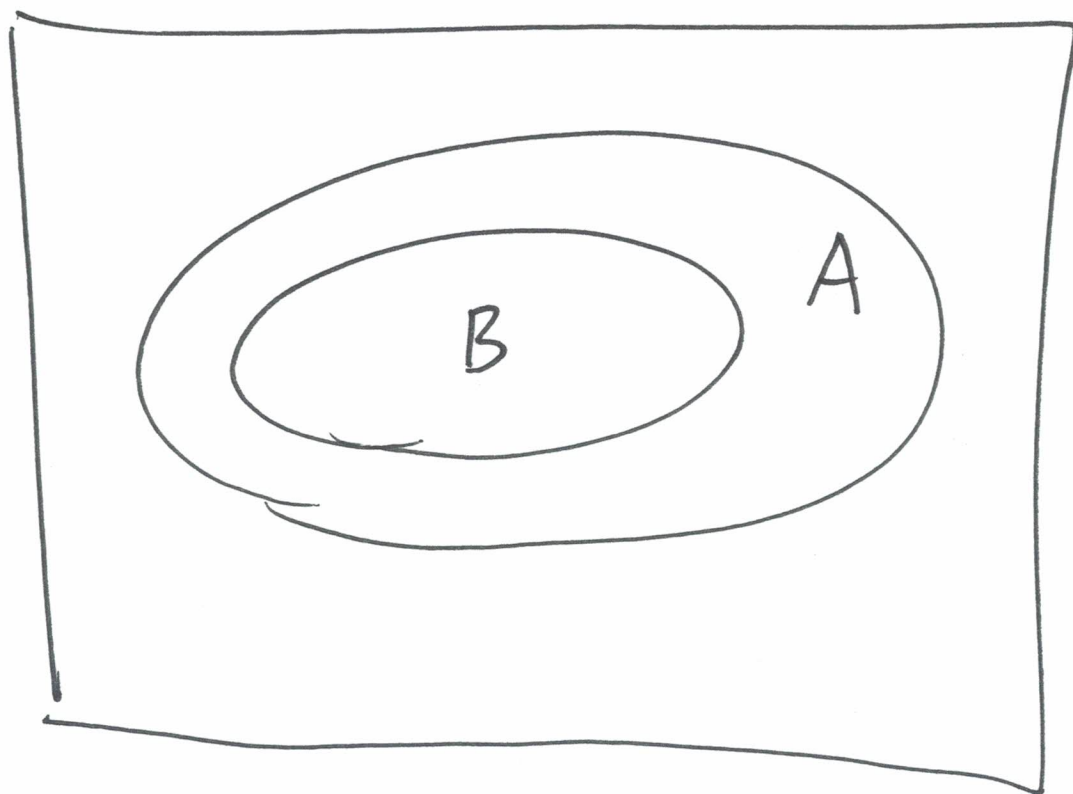




$$P(A) + P(B) - P(A \cap B)$$



$$P(A^c) = 1 - P(A)$$



IF $B \subseteq A$

$$P(B) \leq P(A)$$